Describing Distributions & Linear Regression

ClipSAT Printable Practice Suite (clipsat.org) • Topic 1.03

ANSWER KEY — COMPLETE WORKED SOLUTIONS

Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 **[Foundational] • Target: Describing shape, center, and spread**

A histogram of household incomes has a long tail stretching toward high values, with most households clustered at lower incomes. Describe the shape of this distribution and state which measure of center (mean or median) is more appropriate to report, with justification.

This distribution is skewed right (the long tail points toward high values). The median is more appropriate, since the mean would be pulled upward by the small number of very high incomes, giving a misleadingly high sense of a "typical" household's income; the median is resistant to this effect.

2 **[Foundational] • Target: Describing shape, center, and spread**

Describe what it means for a distribution to be symmetric, and state the relationship between the mean and median in that case.

A symmetric distribution looks approximately the same on both sides of its center when folded in half. In a symmetric distribution, the mean and median are approximately equal, since there is no long tail pulling the mean away from the median.

3 **[Foundational] • Target: Scatterplots and correlation**

The scatterplot shows hours studied (x) versus test score (y) for 8 students, along with the least-squares regression line $\hat{y} = 52 + 6x$.

a Based on the scatterplot, describe the direction and strength of the association between hours studied and test score. The association is positive (as hours studied increases, test score tends to increase) and appears fairly strong, since the points cluster closely around the upward-sloping line.

b Use the regression equation to predict the test score for a student who studies 4.5 hours.
$\hat{y} = 52 + 6(4.5) = 52 + 27 = 79$.

(a) The association is positive (as hours studied increases, test score tends to increase) and appears fairly strong, since the points cluster closely around the upward-sloping line.; (b)
$\hat{y} = 52 + 6(4.5) = 52 + 27 = 79$.

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4 [Foundational] • Target: Interpreting the correlation coefficient

Explain what a correlation coefficient of $r = 0.85$ indicates about a linear relationship, and explain why r alone cannot confirm that x causes y.

$r = 0.85$ indicates a strong, positive linear relationship between the two variables — as one increases, the other tends to increase as well, and the points cluster fairly tightly around a line. However, correlation does not imply causation: a strong linear association could instead be explained by a lurking (confounding) variable influencing both, or simply by coincidence, so r alone cannot establish that changes in x directly cause changes in y.

5 [Foundational] • Target: Interpreting the correlation coefficient

A data set has correlation coefficient $r = -0.4$. Describe the direction and strength of the linear relationship.

The negative sign indicates a negative association (as x increases, y tends to decrease). The magnitude 0.4 indicates a weak-to-moderate linear relationship, since it is much closer to 0 than to ± 1.

6 [Foundational] • Target: Least-squares regression and residuals

Using the regression equation $\hat{y} = 52 + 6x$ from earlier, find the residual for a student who studied 5 hours and actually scored 79.

Predicted: $\hat{y} = 52 + 6(5) = 82$. Residual $=$ observed $-$ predicted $= 79 - 82 = -3$. The negative residual means this student scored 3 points below what the regression line predicted.

7 [Foundational] • Target: Least-squares regression and residuals

Explain what it means for a residual plot to show "no clear pattern," and why this supports using a linear model for the data.

A residual plot with no clear pattern (residuals scattered randomly above and below zero, with roughly constant spread) suggests that a linear model captures the underlying relationship well — any leftover variation is just random scatter, not a systematic trend the line failed to capture (such as a curve). If a residual plot instead showed a clear curve or fan shape, that would suggest a linear model is not appropriate.

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

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8 [Intermediate] • Target: Coefficient of determination

For the study-hours regression, $r = 0.85$. Find r^2 and interpret its meaning in context.

$r^2 = (0.85)^2 = 0.7225$, or about 72.25%. This means approximately 72.25% of the variation in test scores can be explained by the linear relationship with hours studied, leaving about 27.75% of the variation explained by other factors.

9 [Intermediate] • Target: Influential points and outliers in regression

A data point has an unusually large x-value far from the rest of the data, and removing it noticeably changes the slope of the regression line. Explain why this point is called influential, and contrast it with a point that has a large residual but does not change the line much when removed.

A point is influential when its removal substantially changes the regression line (e.g. the slope or intercept) — this typically happens with points that have extreme x-values, since they have high leverage over the line's fit. A point with a large residual but ordinary x-value is simply an outlier in the y-direction; it may look unusual but doesn't have the leverage to substantially shift the line when removed, since points near the middle of the x-range have less influence on slope.

10 [Intermediate] • Target: Extrapolation

Using the regression $\hat{y} = 52 + 6x$ (fit using data from students who studied between 1 and 7 hours), explain why predicting a test score for a student who studied 20 hours would be unreliable.

Predicting for $x = 20$ requires extrapolation — using the regression line far outside the range of x-values (1 to 7 hours) that was actually used to fit it. There is no guarantee the linear relationship continues to hold outside the observed data range (a student's score can't exceed 100, for instance), so such a prediction would likely be unreliable.

11 [Intermediate] • Target: Least-squares regression and residuals

Using the regression model $\hat{y} = 38.0 + 0.55x$, a student who completed 80% of homework scored 86 on the exam. (a) Calculate the residual. (b) Explain what a random scatter of points around 0 on a residual plot indicates about model adequacy.

(a) Residual: Predicted score: $\hat{y} = 38.0 + 0.55(80) = 38.0 + 44.0 = 82.0$. Residual: $e = y - \hat{y} = 86 - 82.0 = \mathbf{+4.0}$ points. The student scored 4.0 points higher than predicted by the regression line. (b) Residual Plot Adequacy: A residual plot showing random scatter with no discernible curve, fanning, or systematic pattern confirms that a linear model is appropriate and that error variance is approximately constant.

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12 **[Intermediate]** • **Target: Coefficient of determination**

A cereal manufacturer places a collectible card in each box: 20% of boxes have Card A, 30% have Card B, and 50% have Card C. Design a simulation using random digits to estimate the probability that a customer must buy at least 6 boxes to collect all three cards.

1. Assign Digits: Assign single digits proportional to probabilities: Digits 0–1 represent Card A (20%); digits 2–4 represent Card B (30%); digits 5–9 represent Card C (50%). 2. Describe One Trial: Read random digits one at a time. Record which card type is obtained on each digit. Continue reading digits until all three card types (A, B, and C) have appeared at least once. Count the total number of digits (boxes) required. 3. Record Statistic: Record whether the total number of boxes opened is ≥ 6 (Yes/No). 4..

13 **[Intermediate]** • **Target: Influential points and outliers in regression**

Level 1 · FoundationsTwo independent events E and F satisfy $P(E) = 0.35$ and $P(F) = 0.60$. Find: (a) $P(E \cap F)$, (b) $P(E \cup F)$, and (c) $P(E \mid F)$.

(a) Intersection: Since E and F are independent: $P(E \cap F) = P(E) \cdot P(F) = 0.35 \times 0.60 = \mathbf{0.21}$. (b) Union: Using the General Addition Rule: $P(E \cup F) = P(E) + P(F) - P(E \cap F) = 0.35 + 0.60 - 0.21 = \mathbf{0.74}$. (c) Conditional: By definition of independence: $P(E \mid F) = P(E) = \mathbf{0.35}$.

14 **[Intermediate]** • **Target: Extrapolation**

Level 2 · MediumLet X and Y be independent random variables with $\sigma_X = 6$ and $\sigma_Y = 8$. What is the standard deviation of the difference $D = X - Y$?

Variances ALWAYS add, even when subtracting random variables:
$\text{Var}(X - Y) = \text{Var}(X) + \text{Var}(Y) = 6^2 + 8^2 = 36 + 64 = 100$. The standard deviation is: $\sigma_{X-Y} = \sqrt{100} = \mathbf{10}$.

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

15 **[Challenging]** • **Target: Describing shape, center, and spread** • **Cross-Lesson: 3.03 The Normal Distribution**

[Unit 3 · Probability, Random Variables & Distributions · Practice 4] If the governing relation for this system is $g(t) = 6t + 6$, evaluate $g(10)$ and find the inverse input t when $g(t) = 72$.

Substitute $t = 10$: $g(10) = 6(10) + 6 = \mathbf{66}$. Setting $6t + 6 = 72$ gives $6t = 66 \Rightarrow t = \mathbf{11}$.

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16 **[Challenging] • Target: Scatterplots and correlation • Cross-Lesson: 3.04 Sampling Distributions & Central Limit Theorem**

Practice 4.2: Bivariate data for study hours x and exam mark y across 10 students yield $\bar{x} = 14.5$, $\bar{y} = 72.0$, and Pearson's correlation coefficient $r = 0.88$. The regression line of y on x has equation $y = 2.4x + c$. (a) Find the value of c. (b) Interpret the slope 2.4 in context. (c) Estimate the exam mark of a student who studies 18 hours.

(a) The regression line must pass through the mean point $(\bar{x}, \bar{y}) = (14.5, 72.0)$: $72.0 = 2.4(14.5) + c \Rightarrow 72.0 = 34.8 + c \Rightarrow c = 72.0 - 34.8 = 37.2$. Thus $y = 2.4x + 37.2$. (b) The slope 2.4 indicates that for each additional hour of study, a student's predicted exam mark increases by 2.4 marks. (c) For $x = 18$: $\hat{y} = 2.4(18) + 37.2 = 43.2 + 37.2 = 80.4$ marks.

17 **[Challenging] • Target: Interpreting the correlation coefficient • Cross-Lesson: 4.01 Confidence Intervals & Hypothesis Testing**

Level 2 · MediumTwo data sets have the same mean of 50 and the same sample size of 100. Dataset A has a standard deviation of 3.2, and Dataset B has a standard deviation of 8.9. Which statement must be true? (A) Dataset A values are more spread out from the mean than Dataset B values. (B) Dataset B has greater variability about the mean than Dataset A. (C) Both datasets have identical minimum and maximum values. (D) Dataset B has more outliers than Dataset A.

Correct Answer: (B). Standard deviation measures the typical distance of data values from their mean. A larger standard deviation directly corresponds to greater variability or dispersion about the mean.

18 **[Challenging] • Target: Least-squares regression and residuals • Cross-Lesson: 4.02 Chi-Square Tests & Regression Inference**

A random sample of 200 students is classified by Grade Level (9th vs 12th) and Smartphone Operating System (iOS vs Android). Observed counts: 9th Grade: 70 iOS, 30 Android (100 total); 12th Grade: 50 iOS, 50 Android (100 total). (a) Calculate the expected count of 9th graders with iOS. (b) Compute the Chi-square test statistic χ^2. (c) With $df = 1$ and $\alpha = 0.05$ (critical value 3.84), conclude whether Grade Level and Operating System are independent.

(a) Expected Count: Total iOS = 120, Total Android = 80, Grand Total = 200. $E_{1,1} = \frac{\text{(Row 1 Total)} \times \text{(Col 1 Total)}}{\text{Grand Total}} = \frac{100 \times 120}{200} = \mathbf{60.0}$. All four expected counts are: 9th/iOS: 60, 9th/Android: 40, 12th/iOS: 60, 12th/Android: 40. (All $E \geq 5$). (b) Chi-Square Statistic: $\chi^2 = \sum \frac{(O-E)^2}{E} = \frac{(70-60)^2}{60} + \frac{(30-40)^2}{40} + \frac{(50-60)^2}{60} + \frac{(50-40)^2}{40}$.

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19 [Challenging] • Target: Coefficient of determination • Cross-Lesson: 1.01 Measures of Center & Spread

In the tomato experiment in Problem 7, the agronomist has two distinct tomato varieties: Roma and Beefsteak. (a) Explain why a randomized block design is superior to a completely randomized design. (b) Outline the blocking protocol.

(a) Superiority of Blocking: Tomato variety is known to have a massive effect on fruit yield. A completely randomized design risks assigning disproportionately more Beefsteak plants to one fertilizer, confounding variety with fertilizer efficacy. Blocking by variety isolates and eliminates the variability between varieties from the experimental error, increasing statistical power. (b) Blocking Protocol: Separate the plants into two blocks: Block 1 (Roma) and Block 2 (Beefsteak). Within Block 1, randomly assign half.

20 [Challenging] • Target: Influential points and outliers in regression • Cross-Lesson: 1.02 Z-Scores & Linear Transformations

A coffee brand claims its jars contain a mean of 200 grams of coffee. A consumer agency takes an SRS of 16 jars, finding $\bar{x} = 197.5$ g and $s = 4.0$ g. Assume jar weights are approximately normally distributed. Conduct a significance test of $H_0 : \mu = 200$ vs. $H_a : \mu < 200$ at $\alpha = 0.05$.

1. State: Test $H_0 : \mu = 200$ g vs $H_a : \mu < 200$ g at $\alpha = 0.05$, where μ is the true mean weight of all coffee jars. 2. Plan: 1-sample t-test for mean. Random sample stated; $16 < 10\%$ of all jars; population stated as approximately normal. 3. Do: $t = \frac{\bar{x} - \mu_0}{s/\sqrt{n}} = \frac{197.5 - 200}{4.0/\sqrt{16}} = \frac{-2.5}{1.0} = -2.50 \quad (df = n - 1 = 15)$. From t-distribution table with $df = 15$: $\text{p} - \text{value} = P(T < -2.50) \approx 0.0123$. 4. Conc.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

21 [Foundational] • Target: Describing shape, center, and spread

A least-squares regression line is $\hat{y} = 16 - 5x$. Predict $\hat{y}$ when $x = 12$.

a. 76

b. -46

c. -42

d. -44 ✓

$\hat{y} = 16 + (-5)(12) = -44$.

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22 [Foundational] • Target: Scatterplots and correlation

A least-squares regression line is $\hat{y} = 52 + 2x$. For a point with $x = 2$ and observed value $y = 49$, find the residual (observed − predicted).

a. -9

b. -5

c. 7

d. -7 ✓

Predicted: $\hat{y} = 52 + (2)(2) = 56$. Residual $= y - \hat{y} = 49 - (56) = -7$.

23 [Foundational] • Target: Interpreting the correlation coefficient

A regression has correlation coefficient $r = -0.85$. Find r^2, expressed as a percentage, to two decimal places.

a. 74.25%

b. 85.00%

c. 70.25%

d. 72.25% ✓

$r^2 = (-0.85)^2 = 0.7225$, or 72.25%.

24 [Foundational] • Target: Least-squares regression and residuals

A distribution has mean 34 and median 47. Based on this comparison, classify the shape of the distribution.

a. Skewed right (mean noticeably greater than median)

b. Cannot be determined from the mean and median alone

c. **Skewed left (mean noticeably less than median)** ✓

d. Approximately symmetric (mean and median are close)

Mean $= 34$, median $= 47$. Skewed left (mean noticeably less than median).

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25 [Foundational] • Target: Coefficient of determination

A regression has coefficient of determination $r^2 = 0.49$. Find the possible value(s) of the correlation coefficient r.

a. ± 0.80

b. ± 0.70 ✓

c. 0.70

d. ± 0.49

$r = \pm\sqrt{0.49} = \pm 0.70$ (since r^2 does not determine the sign of r, both the positive and negative square root are possible).

26 [Foundational] • Target: Influential points and outliers in regression

A social scientist is investigating the relationship between the number of hours spent studying per week (X) and the final exam score (Y) for a group of $n = 100$ students. The least-squares regression line is given by $\hat{y} = 55.2 + 8.5x$. The standard error of the slope is $SE_b = 1.2$. The researcher wants to construct a 95% confidence interval for the true slope of the regression line. What is the value of the t-score used for this confidence interval?

a. 1.645

b. 1.960

c. **1.984** ✓

d. 2.576

The problem asks for the t-score used to construct a 95% confidence interval for the true slope of a regression line. The degrees of freedom for a simple linear regression are $df = n - 2$, where n is the number of data points. In this case, $n = 100$, so $df = 100 - 2 = 98$. We need to find the t-score such that 95% of the area under the t-distribution curve with 98 degrees of freedom is between $-t$ and t. This means that 2.5% of the area is in each tail ($\alpha/2 = 0.025$). Looking up the t-table.

27 [Foundational] • Target: Extrapolation

The correlation coefficient between two variables is $r = -0.31$. What percent of the variation in y is explained by the linear relationship with x?

a. **9.6\%** ✓

b. 0.5\%

c. 19.6\%

d. 29.6\%

$r^2 = (-0.31)^2 = 0.0961$, so about 9.6\% of the variation in y is explained by x.

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28 [Foundational] • Target: Describing shape, center, and spread

The least-squares regression line for a data set is $\hat{y} = 11.1 + 2.5x$. What is the predicted value of y when $x = 45$?

a. 128.6

b. **123.6** ✓

c. 118.6

d. 125.6

$\hat{y} = 11.1 + 2.5(45) = 123.6$.

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 [Intermediate] • Target: Scatterplots and correlation

A scatterplot is shown below. What is the approximate value of the correlation coefficient, r, for the data presented in the scatterplot?

a. -0.9

b. -0.2

c. 0.2

d. **0.9** ✓

The scatterplot shows a strong positive linear association between the two variables. As the values of the x-variable increase, the values of the y-variable also tend to increase. The points cluster closely around an upward-sloping line, indicating a strong positive correlation. Therefore, the correlation coefficient, r, should be close to +1. Among the given choices, 0.9 represents the strongest positive linear relationship.

30 [Intermediate] • Target: Interpreting the correlation coefficient

In a regression analysis with $n = 23$ data points, the slope estimate is $b_1 = 2.31$ with standard error $SE_{b_1} = 0.909$. Find the test statistic t for testing $H_0 : \beta_1 = 0$.

a. 1.04

b. 4.54

c. 1.54

d. **2.54** ✓

$t = \frac{b_1}{SE_{b_1}} = \frac{2.31}{0.909} = 2.54$ with $df = n - 2 = 21$.

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31 [Intermediate] • Target: Least-squares regression and residuals

The coefficient of determination R^2 measures:

a. The slope

b. **Variation in y explained by x ✓**

c. Correlation between residuals

d. Standard error of slope

R^2 = proportion of variation in y explained by the model.

32 [Intermediate] • Target: Coefficient of determination

Correlation coefficient r=–0.9 indicates:

a. Weak negative association

b. Strong positive association

c. **Strong negative association ✓**

d. No association

r close to –1: strong negative linear association.

33 [Intermediate] • Target: Influential points and outliers in regression

Consider the following diagram representing a probability space where events A and B are shown. If $P(A) = 0.6$ and $P(A \cap B) = 0.2$, and the area of the entire sample space is 1, what is the value of $P(B \setminus A)$?

a. 0.2

b. **0.4 ✓**

c. 0.8

d. 0.6

The diagram shows overlapping events A and B within a sample space of area 1. We are given $P(A) = 0.6$ and $P(A \cap B) = 0.2$. We are asked to find $P(B \setminus A)$, which represents the probability of event B occurring but not event A. We know that $P(A) = P(A \cap B) + P(A \cap B^c)$. Since $P(A) = 0.6$ and $P(A \cap B) = 0.2$, we have $0.6 = 0.2 + P(A \cap B^c)$, which means $P(A \cap B^c) = 0.4$. Also, we know that the probability of event B can be expressed as $P(B) = P(A \cap B) + P(B \cap A^c)$. The nota.

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34 [Intermediate] • Target: Extrapolation

A national polling organization is conducting a survey to estimate the proportion of registered voters in a large metropolitan area who support a particular ballot initiative. They plan to use a 95% confidence interval for the population proportion. The organization wants to ensure that the margin of error for this estimate does not exceed 2.5 percentage points. If they anticipate that the sample proportion will be close to 0.5 (the most conservative estimate for planning purposes), what is the minimum sample size, n, they must obtain to achieve this margin of error?

a. 1536

b. **1537** ✓

c. 1535

d. 1534

The formula for the margin of error (ME) of a confidence interval for a population proportion is given by $\text{ME} = z^* \sqrt{\frac{\hat{p}(1-\hat{p})}{n}}$, where z^* is the critical value for the desired confidence level, $\hat{p}$ is the sample proportion, and n is the sample size. For a 95% confidence level, the critical value z^* is approximately 1.96. The desired margin of error is 2.5 percentage points, which is 0.025 in decimal form. To ensure the margin of error does not exceed 0.025, we set up.

35 [Intermediate] • Target: Describing shape, center, and spread

A discrete random variable, X, represents the number of successful trials in a sequence of independent Bernoulli trials. The probability of success on any given trial is $p = 0.6$. The expected value of X is $E[X] = 18$. We are interested in a new random variable, Y, defined as $Y = aX + b$, where a and b are constants. If the variance of Y is $Var(Y) = 7.2$, what is the value of a^2?

a. 0.16

b. **0.36** ✓

c. 1.00

d. 2.50

The random variable X follows a binomial distribution, as it represents the number of successes in a fixed number of independent Bernoulli trials. The expected value of a binomial random variable is given by $E[X] = np$, where n is the number of trials and p is the probability of success. We are given $E[X] = 18$ and $p = 0.6$. Therefore, $np = 18$ implies $n(0.6) = 18$. Solving for n, we get $n = \frac{18}{0.6} = 30$. So, there are 30 trials. The variance of a binomial random variable is given by.

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36 [Intermediate] • Target: Scatterplots and correlation

A bag contains marbles of different colors. The probability of drawing a red marble is $P(\text{Red}) = \frac{2}{5}$ and the probability of drawing a blue marble is $P(\text{Blue}) = \frac{1}{3}$. If the draws are independent, what is the probability of drawing a red marble followed by a blue marble?

a. $\frac{2}{15}$ ✓

b. $\frac{3}{8}$

c. $\frac{1}{2}$

d. $\frac{11}{15}$

Since the draws are independent, the probability of two independent events occurring in sequence is the product of their individual probabilities. Let R be the event of drawing a red marble and B be the event of drawing a blue marble. We are given: $P(R) = \frac{2}{5}$ $P(B) = \frac{1}{3}$ The probability of drawing a red marble followed by a blue marble is $P(R \text{ and } B) = P(R) \times P(B)$. $P(R \text{ and } B) = \frac{2}{5} \times \frac{1}{3} = \frac{2 \times 1}{5 \times 3} = \frac{2}{15}$. Therefo.

37 [Intermediate] • Target: Interpreting the correlation coefficient

A discrete random variable X can take on the values $1, 2, 3,$ and 4. The probability mass function $P(X = x)$ is given by $P(X = x) = kx$ for $x \in \{1, 2, 3, 4\}$. What is the value of k?

a. **1/10** ✓

b. 1/4

c. 1/6

d. 1/16

The sum of probabilities for all possible values of a discrete random variable must equal 1. Therefore, $\sum_{x=1}^{4} P(X = x) = 1$. Substituting the given probability mass function, we have: $P(X = 1) + P(X = 2) + P(X = 3) + P(X = 4) = 1$ $k(1) + k(2) + k(3) + k(4) = 1$ Factoring out k: $k(1 + 2 + 3 + 4) = 1$ $k(10) = 1$ Solving for k: $k = \frac{1}{10}$. Common algebraic traps include incorrectly summing the values of x or forgetting to set the sum of probabilities to 1. For example, setting $k = 1$ or $k = 1/4$ wo.

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

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38 [Challenging] • Target: Least-squares regression and residuals • Cross-Lesson: 4.02 Chi-Square Tests & Regression Inference

A data set consists of the following five points: $P_1 = (2, 3), P_2 = (4, y), P_3 = (6, 7), P_4 = (8, 9), P_5 = (10, 11)$. If the mean of the x-coordinates is 6 and the mean of the y-coordinates is 7, what is the value of y?

a. 3

b. 4

c. **5** ✓

d. 6

The mean of the x-coordinates is given by $\frac{2+4+6+8+10}{5} = \frac{30}{5} = 6$. This matches the information given in the problem. The mean of the y-coordinates is given by $\frac{3+y+7+9+11}{5}$. We are told that the mean of the y-coordinates is 7. Therefore, we can set up the equation: $\frac{3+y+7+9+11}{5} = 7$ Combine the constant terms in the numerator: $\frac{30+y}{5} = 7$ Multiply both sides by 5: $30 + y = 35$ Subtract 30 from both sides: $y = 5$ Therefore, the value of y.

39 [Challenging] • Target: Coefficient of determination • Cross-Lesson: 1.01 Measures of Center & Spread

A fair six-sided die is rolled. What is the probability of rolling a number greater than 4?

a. 1/6

b. **1/3** ✓

c. 1/2

d. 2/3

The possible outcomes when rolling a fair six-sided die are {1, 2, 3, 4, 5, 6}. The total number of possible outcomes is 6. The outcomes that are greater than 4 are {5, 6}. The number of favorable outcomes is 2. The probability of an event is the ratio of the number of favorable outcomes to the total number of possible outcomes. Therefore, the probability of rolling a number greater than 4 is $\frac{2}{6} = \frac{1}{3}$.

40 [Challenging] • Target: Influential points and outliers in regression • Cross-Lesson: 1.02 Z-Scores & Linear Transformations

Expected value E(X) for discrete X:

a. The most frequent value

b. $\sum x_i P(x_i)$ ✓

c. The median

d. $n \cdot p$ always

Applying Unit 3 · Probability, Random Variables & Distributions principles, Weighted average of outcomes. Therefore, the verified correct option is $\sum x_i P(x_i)$.

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41 [Challenging] • Target: Extrapolation • Cross-Lesson: 2.01 Sampling Methods & Bias

If P(A)=0.4 and P(B|A)=0.5, then P(A∩B)=

a. 0.2 ✓

b. 0.5

c. 0.4

d. 0.9

Applying Unit 3 · Probability, Random Variables & Distributions principles, P(A∩B)=P(A)P(B|A)=0.4×0.5=0.2. Therefore, the verified correct option is 0.2.

42 [Challenging] • Target: Describing shape, center, and spread • Cross-Lesson: 2.02 Experimental Design & Scope of Inference

A binomial random variable has $n = 25$ and $p = 0.64$. Find its mean $\mu = np$.

a. 13

b. 14

c. 16 ✓

d. 18

Applying Unit 3 · Probability, Random Variables & Distributions principles, $\mu = np = 25 \times 0.64 = 16$. Therefore, the verified correct option is 16.

43 [Challenging] • Target: Scatterplots and correlation • Cross-Lesson: 3.01 Probability Rules & Conditional Probability

If $P(A) = 0.56$, find $P(A^c)$ (the complement of A).

a. 0.44 ✓

b. 0.54

c. 0.64

d. 0.24

Applying Unit 3 · Probability, Random Variables & Distributions principles, $P(A^c) = 1 - P(A) = 1 - 0.56 = 0.44$. Therefore, the verified correct option is 0.44.

44 [Challenging] • Target: Interpreting the correlation coefficient • Cross-Lesson: 3.02 Discrete Random Variables & Binomial Distributions

A geometric random variable has success probability $p = 0.48$. Find the mean (expected number of trials until first success), $\mu = 1/p$.

a. 2.08 ✓

b. 8.08

c. 4.08

d. 5.08

Applying Unit 3 · Probability, Random Variables & Distributions principles, $\mu = 1/p = 1/0.48 = 2.08$. Therefore, the verified correct option is 2.08.

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45 [Challenging] • Target: Least-squares regression and residuals • Cross-Lesson: 3.03 The Normal Distribution

In a bivariate regression with sample size $n = 31$ (Research Model 1), sample slope is $b_1 = 1.70$ with $\text{SE}_{b_1} = 0.26$. Testing $H_0 : \beta_1 = 0$, what is the t-test statistic and degrees of freedom?

a. $t = 6.54$, $\text{df} = 29$ ✓

b. $t = 0.44$, $\text{df} = 29$

c. $t = 8.08$, $\text{df} = 31$

d. $t = 6.54$, $\text{df} = 30$

$t = \frac{b_1 - 0}{\text{SE}_{b_1}} = \frac{1.70}{0.26} = 6.54$, with $\text{df} = n - 2 = 31 - 2 = 29$.

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